

# HAITAO XU

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## EDUCATION

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**Cleveland State University, Cleveland, OH** Fall 2021 - Summer 2025  
Ph.D. in Computer Science GPA: 3.84/4.00  
Dissertation: Algorithms for Facility Location Problems with Uncertainty on Graphs

**Kunming University of Science and Technology, Kunming, China** Fall 2011 - Summer 2014  
M.S. in Software Engineering GPA: 85.7/100  
Thesis: Research and Application of Software Component Technology in Examination System

**Shandong Institute of Commerce and Technology, Jinan, China** Fall 2005 - Summer 2008  
A.S. in Computer Science and Applications GPA: 82.3/100

## RESEARCH INTERESTS

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Algorithms and Data Structures, Combinatorial Optimization, Operations Research, Theoretical Computer Science

## PUBLICATIONS

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### Peer Reviewed Journal Articles

1. **Haitao Xu** and Jingru Zhang. The k-Center Problem of Uncertain Points on Graphs. *Submitted to the Information Processing Letters*.
2. **Haitao Xu** and Jingru Zhang. The two-center problem of uncertain points on cactus graphs. *Journal of Combinatorial Optimization*, 49.4 (2025): 54.
3. **Haitao Xu** and Jingru Zhang. The two-center problem of uncertain points on a real line. *Journal of Combinatorial Optimization*, 45.2 (2023): 68.
4. Qi Sun, Liwen Jiang, **Haitao Xu**. A Double-Layer Combination Algorithm for Real-Time Information-Sharing Network Design Problem. *Complexity*, 2021(1), 4856593.
5. Qi Sun, Liwen Jiang, **Haitao Xu**. Expectation-Maximization Algorithm of Gaussian Mixture Model for Vehicle-Commodity Matching in Logistics Supply Chain. *Complexity*, 2021(1), 9305890.
6. Song, Yang, Yan-Qiu Liu, Qi Sun, Ming-Fei Chen, and **Hai-Tao Xu**. A Joint Optimization Model considering the Product User's Risk Preference for Supply System Disruption. *Mathematical Problems in Engineering*, no. 1 (2021): 5081753.

### Peer Reviewed Conference Papers

1. **Haitao Xu** and Jingru Zhang. The two-center problem of uncertain points on trees. *In Proc. of the 16th Annual International Conference on Combinatorial Optimization and Applications (COCO A)*, pages 485-697, 2023.
2. Qi Sun, **Haitao Xu**. Adaptive Control Method of UAV Intelligent Rudder Based on Hybrid Genetic Algorithm. *In Proc. of the 33rd Chinese Control and Decision Conference (CCDC)*, pp. 4753-4757. IEEE, 2021.
3. Lijie Wu, Qi Sun, **Haitao Xu**, Xiaochen Song, Yang Zhang. Design of hybrid simulated annealing algorithm for UAV scheduling based on coordinated task scheduling. *In Proc. of the 40th Chinese Control Conference (CCC)*, pages 1669-1674. IEEE, 2021.

## PROFESSIONAL EXPERIENCE

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### **Cleveland State University, Cleveland, Ohio**

Fall 2021 - Present

*Research Assistant:* We study several facility location problems on various graphs related to uncertain demand data. Specifically, we focus on solving the two-center problem on a line, a tree graph, and a cactus graph, as well as the k-center problem on a general graph. These center problems are classical combinatorial optimization problems and have been explored a lot in theoretical computer science. Our emphasis, however, lies in uncertain demand data, leading us to develop more general algorithmic approaches to address these problems.

*Teaching Assistant:* CIS550 Introduction to Algorithms; CIS606 Analysis of Algorithms; CIS620 Advanced Operating Systems

### **KEDACOM, Shanghai, China**

Apr 2016 - Feb 2020

*Software Engineer*

Participated in the development of core components for a cross-platform video conferencing application supporting Windows PC, Android, and iOS. Key responsibilities included API encapsulation, data transmission, command dispatching, and real-time messaging based on the XMPP protocol. Also served as a technical trainer for new team members by introducing development workflows and system architecture to help them onboard quickly.

### **Shanghai Huateng Software Systems Co., Ltd, Shanghai, China**

Aug 2014 - Apr 2016

*Software Engineer*

Participated in the development of a high-reliability, high-concurrency banking transaction system, including features such as information transmission, message distribution, batch transaction settlement, card services, and card security authentication. Additionally involved in deploying new modules to the bank's backend production servers.

### **Jinan Yuancheng Innovation Technology Co., Ltd, Jinan, China**

July 2008 - June 2010

*Software Engineer*

Participated in the development of an automated office system using Java, Ext JS, JavaScript, CSS, and Windows Workflow Foundation, primarily designed to support process review and approval workflows.

## TECHNICAL SKILLS

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1. Over 5 years of software development experience with a solid foundation in software analysis, design, system programming, and maintenance.
2. Experienced in software design and development using C/C++, Java, and Python, with hands-on experience in multithreading, data structures, algorithms, socket programming, and design patterns.
3. Practical experience in Oracle and MySQL programming, including writing high-performance SQL and performing database optimization and tuning.
4. Experience in web development using Java, JavaScript, HTML, and CSS.

## PATENTS

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1. Asynchronous multi-process service processing system, method, apparatus, and storage medium; No. CN109032767A

## SERVICES

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### **Journal Reviewer & Conference Reviewer**

Hindawi Complexity; The 40th Chinese Control Conference (CCC'2021);